

Screenshots of 5 layer protocol headers from Wireshark:

Request(<http://cspb.teletalk.com.bd/>)

Layer1:

```
▼ Frame 48: Packet, 523 bytes on wire (4184 bits), 523 bytes captured (4184 bits) on interface \Device\NPF_{83B2968E-57C0-4A73-B5A4-EE38ED7DC4E0}, id 0
  Section number: 1
  ▸ Interface id: 0 (\Device\NPF_{83B2968E-57C0-4A73-B5A4-EE38ED7DC4E0})
    Encapsulation type: Ethernet (1)
    Arrival Time: Jul  9, 2026 10:44:00.376554000 Bangladesh Standard Time
    UTC Arrival Time: Jul  9, 2026 04:44:00.376554000 UTC
    Epoch Arrival Time: 1783572240.376554000
    [Time shift for this packet: 0.00000000 seconds]
    [Time delta from previous captured frame: 657.000 microseconds]
    [Time since reference or first frame: 2.584827300 seconds]
    Frame Number: 48
    Frame Length: 523 bytes (4184 bits)
    Capture Length: 523 bytes (4184 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

Screenshot shows the packet is 523 bytes, arrival timestamp, and reveals that it contains an HTTP request layered over Ethernet, IP, and TCP protocols.

Layer2:

```
▼ Ethernet II, Src: MicroStarINT_4d:95:bc (d8:43:ae:4d:95:bc), Dst: TpLinkTechno_20:98:18 (ac:84:c6:20:98:18)
  ▸ Destination: TpLinkTechno_20:98:18 (ac:84:c6:20:98:18)
    .... 00. .... = LG bit: Globally unique address (factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  ▸ Source: MicroStarINT_4d:95:bc (d8:43:ae:4d:95:bc)
    .... 00. .... = LG bit: Globally unique address (factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

This is the Data Link layer, and the screenshot shows the source and destination MAC addresses, and identifies the payload as an IPv4 packet.

Layer3:

```
▼ Internet Protocol Version 4, Src: 192.168.0.104, Dst: 103.230.106.216
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 509
  Identification: 0xa195 (41365)
  ▸ 010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.104
  Destination Address: 103.230.106.216
  [Stream index: 10]
```

This is the Network layer (IPv4), the screenshot shows the source IP (192.168.0.104, my device) and destination IP (103.230.106.216, the web server) used to route the packet across the internet, along with fields like TTL (128) and Protocol (TCP), indicating the transport layer above it will use TCP.

Layer4:

```
▼ Transmission Control Protocol, Src Port: 52047, Dst Port: 80, Seq: 1, Ack: 1, Len: 469
  Source Port: 52047
  Destination Port: 80
  [Stream index: 5]
  [Stream Packet Number: 1]
  ▶ [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 469]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 4245531422
  [Next Sequence Number: 470 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 2926677978
  0101 .... = Header Length: 20 bytes (5)
  ▶ Flags: 0x018 (PSH, ACK)
  Window: 64385
  [Calculated window size: 64385]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0x95be [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▶ [Timestamps]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (469 bytes)
```

This is the Transport layer (TCP), the screenshot shows the source port (52047) and destination port (80, standard HTTP port).

Layer5:

```
▼ Hypertext Transfer Protocol
  ▶ GET / HTTP/1.1\r\n
  Host: cspb.teletalk.com.bd\r\n
  Connection: keep-alive\r\n
  Upgrade-Insecure-Requests: 1\r\n
  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/149.0.0.0 Safari/537.36\r\n
  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
  Referer: https://www.google.com/\r\n
  Accept-Encoding: gzip, deflate\r\n
  Accept-Language: en-US,en;q=0.9\r\n
  \r\n
  [Full request URI: http://cspb.teletalk.com.bd/]
```

This is the Application layer (HTTP), the screenshot shows the actual HTTP GET request targeting cspb.teletalk.com.bd/.

Response

Layer1:

```
▼ Frame 182: Packet, 438 bytes on wire (3504 bits), 438 bytes captured (3504 bits) on interface \Device\NPF_{83B2968E-57C0-4A73-B5A4-EE38ED7DC4E0}, id 0
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{83B2968E-57C0-4A73-B5A4-EE38ED7DC4E0})
    Interface name: \Device\NPF_{83B2968E-57C0-4A73-B5A4-EE38ED7DC4E0}
    Interface description: Ethernet
    Encapsulation type: Ethernet (1)
    Arrival Time: Jul  9, 2026 10:44:01.442610300 Bangladesh Standard Time
    UTC Arrival Time: Jul  9, 2026 04:44:01.442610300 UTC
    Epoch Arrival Time: 1783572241.442610300
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 893.400 microseconds]
    [Time since reference or first frame: 3.650883600 seconds]
    Frame Number: 182
    Frame Length: 438 bytes (3504 bits)
    Capture Length: 438 bytes (3504 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

This is the Frame layer of the HTTP response packet (Frame 182), the screenshot shows this is a 438-byte frame captured shortly after the request .

Layer2:

```
▼ Ethernet II, Src: TpLinkTechno_20:98:18 (ac:84:c6:20:98:18), Dst: MicroStarINT_4d:95:bc (d8:43:ae:4d:95:bc)
  ▼ Destination: MicroStarINT_4d:95:bc (d8:43:ae:4d:95:bc)
    .... 00. .... = LG bit: Globally unique address (factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  ▼ Source: TpLinkTechno_20:98:18 (ac:84:c6:20:98:18)
    .... 00. .... = LG bit: Globally unique address (factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

This is the Data Link layer of the response packet, and the source and destination MAC addresses are now reversed.

Layer3:

```
▼ Internet Protocol Version 4, Src: 103.230.106.216, Dst: 192.168.0.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... 00.. = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 424
  Identification: 0x9d24 (40228)
  ▼ 010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 245
  Protocol: TCP (6)
  Header Checksum: 0x535c [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 103.230.106.216
  Destination Address: 192.168.0.104
  [Stream index: 10]
```

This is the Network layer (IPv4) of the response packet — the source and destination IPs are now reversed (103.230.106.216 → 192.168.0.104), confirming the web server is sending this packet back to your device over TCP.

Layer4:

```
▼ Transmission Control Protocol, Src Port: 80, Dst Port: 52047, Seq: 4049, Ack: 824, Len: 384
  Source Port: 80
  Destination Port: 52047
  [Stream index: 5]
  [Stream Packet Number: 14]
  ▶ [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 384]
  Sequence Number: 4049 (relative sequence number)
  Sequence Number (raw): 2926682026
  [Next Sequence Number: 4433 (relative sequence number)]
  Acknowledgment Number: 824 (relative ack number)
  Acknowledgment number (raw): 4245532245
  0101 .... = Header Length: 20 bytes (5)
  ▶ Flags: 0x018 (PSH, ACK)
  Window: 22984
  [Calculated window size: 22984]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0xc77a [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▶ [Timestamps]
  ▶ [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (384 bytes)
```

This is the Transport layer (TCP) of the response packet, the source port is now 80 (web server) and destination port is 52047 (my browser).

Layer5:

```
▼ Hypertext Transfer Protocol, has 2 chunks (including last chunk)
  ▶ HTTP/1.1 200 OK\r\n
    Server: nginx\r\n
    Date: Thu, 09 Jul 2026 04:42:55 GMT\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    Transfer-Encoding: chunked\r\n
    Connection: keep-alive\r\n
    Vary: Accept-Encoding\r\n
    X-Powered-By: PHP/5.6.40\r\n
    X-Frame-Options: SAMEORIGIN\r\n
    X-XSS-Protection: 1; mode=block\r\n
    Content-Encoding: gzip\r\n
    \r\n
    [Request in frame: 180]
    [Time since request: 7.860400 milliseconds]
    [Request URI: /time.php]
    [Full request URI: http://cspb.teletalk.com.bd/time.php]
  ▶ HTTP chunked response
    Content-encoded entity body (gzip): 75 bytes -> 80 bytes
    File Data: 80 bytes
```

This is the Application layer (HTTP) of the response, screenshot shows the server's HTTP/1.1 200 OK reply, confirming the request succeeded.